

The Latest Trends in the Programming World



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The beginning of a new year is when one takes stock of things. In this article, the author uses four popular ranking schemes to arrive at the top ten programming languages of 2021.

The four most popular ranking schemes for programming languages are the TIOBE Index, the RedMonk Programming Language Rankings, the PYPL (Popularity of Programming Language) Index, and the IEEE Spectrum Ranking of Programming Languages. I have used these to find out the 10 most popular programming languages in 2021. Five of the top 10 languages in all the four rankings are the same. These are: Java, C, C++, Python, and JavaScript. For the first time, C# and PHP are not in the top 10 list of one of the rankings, the IEEE Spectrum Ranking of Programming Languages. A modest 14th rank by PHP and a poor 23rd rank by C# in the IEEE Spectrum Ranking was indeed surprising. Along with PHP and C#, R programming language

also has only three top 10 finishes in the four rankings. R is placed 13th in the RedMonk Programming Language ranking but has improved ratings in all the four rankings. TypeScript and Swift have two top 10 finishes. So, our top 10 list of popular programming languages for 2021 is the following:

- Java
- C
- C++
- Python
- JavaScript
- PHP
- C#
- R
- TypeScript
- Swift

From our top 10 list for 2019, Objective-C has been replaced by TypeScript in 2021. Notice that there is a parallel scripting language, also

called Swift, but the one included in our top 10 list is Swift developed by Apple Inc. Also note that Visual Basic, Groovy, Ruby, CSS, Arduino language, Go, MATLAB, and Objective-C all have one top 10 finish each. Groovy is ranked 10th in the TIOBE Index but in two other rankings it is ranked 25th and 40th, and in a fourth ranking it couldn't make it to the top 20. This clearly indicates how the individual biases by various ranking schemes can affect the final rank of a programming language, except maybe of the very popular programming languages like C, C++, Java, Python, etc.

The programming languages race: Winners and losers

Now let us see which programming languages are steadily becoming popular. Previously, in 2019, I had